

# Cardamo: An AI-Based Smart Cardamom Cultivation Management System Using Computer Vision and Predictive Analytics

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**Abstract**—Recent advancements in computer vision, machine learning, and predictive analytics have improved smart agriculture systems by enabling automated crop monitoring, quality assessment, and data-driven decision-making. This paper presents Cardamo, an ML-driven smart cardamom cultivation management system designed to support Sri Lankan cardamom farmers in disease detection, pod monitoring, quality grading, and market decision support. The system consists of four primary components: cardamom leaf disease detection, smart pod monitoring and caring, cardamom pod quality grading, and market price prediction with profit optimization. The leaf disease detection component applies image preprocessing and deep learning-based classification to identify common cardamom leaf diseases such as leaf blight, leaf spot, mosaic disease, chlorotic streak, and vein clearing. The pod monitoring component applies EfficientNetB0-based image classification to identify pod maturity and pod-related conditions such as capsule borer damage, thrips damage, capsule rot, pod splitting, and shrivelling. The pod quality grading component analyzes harvested cardamom pod images based on visual features such as size, color, shape, and surface defects to classify pods into suitable quality grades. Explainable AI techniques are integrated to provide visual explanations for disease

detection and grading decisions, improving transparency and user trust. The market prediction component analyzes historical price trends, seasonal production patterns, exchange rate variations, processing costs, and fresh-to-dried weight conversion ratios to generate profit-based “Hold or Sell” recommendations. Through this integrated pipeline, Cardamo delivers a user-friendly and practical decision-support solution that enhances crop health monitoring, reduces yield losses, improves export-quality production, and supports profitable cardamom cultivation.

**Index Terms**—Smart Agriculture, Cardamom Cultivation, Machine Learning, Computer Vision, Deep Learning, Explainable AI, Quality Grading, Market Price Prediction

## I. INTRODUCTION

Over the past years, the application of computer vision, machine learning, deep learning, and predictive analytics has transformed the agricultural sector by introducing intelligent tools for crop monitoring, disease identification, quality assessment, and market-based decision-making. These technologies have improved traditional farming

practices by enabling automated analysis of crop images, early detection of plant health issues, and data-driven recommendations for farmers. In smart agriculture, such systems are especially important for high-value crops where product quality, disease control, harvesting decisions, and market timing directly affect farmer income and export performance.

Cardamom, commonly known as the “Queen of Spices,” is one of the most valuable spice crops cultivated in Sri Lanka and plays an important role in the national spice export economy [1], [2]. It is mainly grown in humid highland regions such as Kandy, Matale, and Badulla, where environmental conditions strongly influence crop health, pod development, and final product quality. However, cardamom cultivation faces several practical challenges, including leaf diseases, pod-related disorders, inaccurate maturity identification, inconsistent quality grading, and market price fluctuations. Previous studies on cardamom cultivation and agroecosystems have shown that pests, diseases, environmental conditions, and post-harvest handling can significantly affect crop yield, quality, and long-term sustainability [3], [4].

One of the major challenges faced by cardamom farmers is the early identification of leaf diseases. Common diseases such as leaf blight, leaf spot, mosaic disease, chlorotic streak, and vein clearing can spread rapidly in plantation environments and reduce crop productivity. Traditionally, farmers identify these diseases through manual observation, which can be time-consuming, subjective, and inaccurate, especially when symptoms appear similar across different diseases. Deep learning-based image classification has shown strong performance in plant disease detection, where convolutional neural networks are used to identify disease patterns from leaf images [5], [6]. Foundational studies in deep learning and CNN-based image recognition have further demonstrated the effectiveness of automated visual feature extraction for classification tasks [7], [8]. However, many existing systems focus on general crops or laboratory image datasets, while limited research has specifically addressed cardamom leaf disease detection under real Sri Lankan plantation conditions.

To address this requirement, the proposed system uses EfficientNetB0 as the image classification model for cardamom leaf disease detection. EfficientNet

introduces a compound scaling method that balances network depth, width, and input resolution to improve both accuracy and computational efficiency [30]. Because EfficientNetB0 is the baseline and lightweight version of the EfficientNet family, it is suitable for image-based agricultural applications that require efficient deployment in web or mobile-supported environments. In the proposed system, cardamom leaf images are preprocessed using resizing, normalization, and noise reduction before being classified into disease categories. This enables the system to support early disease detection and assist farmers in taking timely crop management actions.

In addition to leaf diseases, pod monitoring is important because the cardamom pod is the main commercial product of the plant. Pod-related issues such as capsule borer damage, thrips damage, capsule rot, pod splitting, shrivelling, and incorrect maturity identification can reduce both yield and market value. Manual pod inspection is difficult due to changes in lighting, background noise, pod color variation, and the similarity between some healthy and unhealthy pod appearances. Therefore, image-based deep learning can support more consistent identification of pod conditions. In Cardamo, EfficientNetB0 is also applied for pod monitoring by learning visual patterns such as pod color, shape, surface texture, maturity appearance, and visible damage indicators. This allows the system to classify pod conditions and support better harvesting and caring decisions.

Cardamom pod quality grading is another major challenge in the industry. At present, grading is commonly performed manually based on visible characteristics such as size, color, shape, surface quality, and defects. This manual process can be subjective and may vary depending on the experience of the grader. Previous research has explored computer vision and machine learning techniques for cardamom grading, capsule size estimation, and sorting automation [14]–[18]. Other agricultural quality assessment studies have also shown that CNN-based image classification can successfully identify quality-related visual patterns in agricultural products [22], [23]. In the proposed system, a Keras-based CNN model is used for pod quality grading to classify harvested cardamom pods into suitable quality grades based on visual features. Explainable Artificial Intelligence techniques such as Grad-CAM can be used to visually highlight the regions that influenced

the model decision, improving transparency and user trust [10].

Apart from crop health and quality, cardamom farmers also face market-related challenges. A major post-harvest decision is whether to sell fresh cardamom pods immediately or dry them and sell later for a potentially higher price. This decision is influenced by historical price trends, seasonal production patterns, exchange rate fluctuations, drying costs, labor costs, and the fresh-to-dried weight conversion ratio. Previous studies have shown that agricultural export prices can be affected by currency fluctuations and value addition costs in the spice sector [24], [28]. Time-series forecasting and machine learning-based prediction methods have also been applied in agriculture to support market planning and decision-making [25], [27], [29]. However, many forecasting approaches do not provide localized decision support for Sri Lankan cardamom farmers by comparing fresh-selling and dried-selling profitability.

For the market analysis component, the proposed system uses Ridge Regression to support price prediction and profit optimization. Ridge Regression is useful when multiple input variables are involved because it applies regularization to reduce overfitting and improve prediction stability in regression tasks [31]. In Cardamo, historical cardamom prices, exchange rate variations, seasonal trends, drying costs, labor costs, and processing expenses are considered as input features. The model supports the prediction of market price behavior, while the profit optimization engine compares the expected income from selling fresh pods with the expected income from drying and selling later. Based on this comparison, the system generates a practical “Hold or Sell” recommendation for farmers.

To address these challenges, this study introduces Cardamo, an ML-driven smart cardamom cultivation management system designed to support Sri Lankan cardamom farmers through an integrated digital platform. The system consists of four main components: cardamom leaf disease detection, smart pod monitoring, cardamom pod quality grading, and market price prediction with profit optimization. The leaf disease detection and pod monitoring components apply image preprocessing and EfficientNetB0-based classification, while the pod quality grading component uses a Keras-based CNN model. The

market prediction component applies Ridge Regression to analyze historical prices, exchange rates, drying costs, and fresh-to-dried conversion factors to provide profit-based recommendations.

This integrated system addresses the current lack of a complete AI-based decision-support platform for cardamom cultivation in Sri Lanka. By combining EfficientNetB0-based image classification with Ridge Regression-based market analysis, Cardamo supports early disease identification, pod condition monitoring, consistent quality grading, and data-driven selling decisions. Through this approach, the proposed system aims to reduce crop losses, improve export-quality production, minimize manual errors, and support more profitable and sustainable cardamom cultivation.

## II. LITERATURE REVIEW

In recent years, the integration of computer vision, deep learning, and predictive analytics has become increasingly important in smart agriculture for improving crop monitoring, disease detection, quality grading, and market-based decision-making. A smart cardamom cultivation management system such as Cardamo requires the combined use of image classification, preprocessing techniques, interpretable deep learning, and predictive modelling. This literature review positions each component of the proposed system within the current technological context by reviewing related work in cardamom cultivation, plant disease detection, pod monitoring, quality grading, and agricultural market prediction.

Cardamom is one of the most valuable spice crops in Sri Lanka and contributes significantly to the national spice export sector [1], [2]. It is mainly cultivated in highland regions such as Kandy, Matale, and Badulla, where environmental conditions strongly influence crop productivity and quality. Studies on cardamom cultivation and agroecosystems highlight that pests, diseases, soil conditions, and post-harvest handling practices directly affect yield, sustainability, and export value [3], [4]. These findings show the need for intelligent agricultural systems that can support farmers through automated monitoring and decision-making.

Leaf disease detection is a major area in which deep learning has shown strong potential. Traditional disease identification is usually performed through

manual inspection by farmers or agricultural officers, which can be time-consuming and subjective. Previous studies have demonstrated that image-based plant disease detection using convolutional neural networks can identify disease symptoms such as discoloration, spots, and texture changes from leaf images [5], [6]. Foundational deep learning studies have further shown the effectiveness of CNN-based architectures in extracting complex visual features from image data [7], [8]. In addition, few-shot learning techniques have been explored to improve model performance when only a limited number of labelled images are available [9]. These studies provide the foundation for applying deep learning to cardamom leaf disease detection.

For the leaf disease detection component of Cardamo, EfficientNetB0 is selected as the main image classification model. EfficientNet-based architectures are designed to achieve better accuracy and computational efficiency by scaling network depth, width, and resolution in a balanced manner. This makes EfficientNetB0 suitable for agricultural image classification tasks, especially when the system needs to perform efficiently in a web or mobile-based environment. In the proposed system, cardamom leaf images are first preprocessed through resizing, normalization, and noise reduction, and then passed into the EfficientNetB0 model to classify common cardamom leaf conditions such as healthy leaves, leaf blight, leaf spot, mosaic disease, chlorotic streak, and vein clearing. This approach supports early disease identification and helps farmers take timely action to reduce crop losses.

Pod monitoring is another important component because the cardamom pod is the main commercial product of the plant. Pod-related issues such as capsule borer damage, thrips damage, capsule rot, pod splitting, shrivelling, and incorrect maturity identification can reduce both yield and market value. Manual pod monitoring is difficult because visual symptoms may vary depending on pod maturity, lighting conditions, background noise, and field environment. Instead of relying only on manual inspection, deep learning-based image classification can support more consistent pod condition identification. In the proposed system, EfficientNetB0 is used for pod monitoring because it can extract important visual patterns such as pod color, surface texture, shape variation, and visible defect patterns. By applying this model to pod images, the system can

classify pod conditions and support farmers in identifying maturity or damage-related issues.

Automated cardamom pod quality grading has also been studied using computer vision and machine learning techniques. Traditional grading depends on visible features such as pod size, color, shape, texture, and surface defects. However, this process is subjective and may vary between graders. Previous studies on cardamom grading have shown that computer vision can be used to classify pods into different quality categories [14], while other research has combined computer vision with additional sensing methods to improve grading accuracy [15]. Image processing approaches have also been used to estimate cardamom capsule size, surface area, and physical characteristics, showing the importance of geometric features in grading [16]. Machine learning-based grading and sorting systems have further demonstrated the possibility of automating cardamom quality assessment [17], [18]. These studies support the development of a deep learning-based grading component in Cardamo.

For the pod quality grading component, Cardamo uses a Keras-based CNN model to classify harvested cardamom pods into quality grades based on visual appearance. The model analyzes features such as color uniformity, pod shape, surface quality, and visible defects. Related agricultural studies have shown that CNN-based image classification and defect detection approaches can successfully identify quality-related visual patterns in agricultural products such as fruits and spices [22], [23]. By using EfficientNetB0, the grading component aims to provide a lightweight and accurate classification approach that can support farmers, traders, and exporters in making more consistent quality assessments. To improve transparency, explainable AI techniques such as Grad-CAM can be integrated to highlight image regions that influenced the model's grading decision [10]. This helps users understand whether the system focused on relevant pod features such as discoloration, surface damage, or shape irregularities.

Market prediction and profit optimization form the fourth major component of the Cardamo system. Cardamom farmers often face uncertainty when deciding whether to sell fresh pods immediately or dry them and sell later. This decision depends on historical price trends, seasonal production, currency exchange

rates, drying costs, labor costs, and the fresh-to-dried conversion ratio. Studies on agricultural export commodity prices in Sri Lanka show that currency fluctuations can influence local market prices [24]. Time-series forecasting and machine learning-based prediction methods have also been applied to agricultural commodity price analysis to support market planning [25], [27], [29]. Moreover, value addition and processing cost studies in the spice sector emphasize that drying costs, fuel costs, labor, and post-harvest handling expenses must be considered when evaluating profitability [28].

For the market analysis component, Ridge Regression is used as the predictive model. Ridge Regression is suitable for market prediction tasks because it can handle multiple input variables while reducing overfitting through regularization. In the proposed system, factors such as historical cardamom prices, exchange rate changes, seasonal trends, drying costs, and processing expenses are used as input features. The model predicts possible market price behavior and supports the profit optimization engine. The system then compares the expected profit from selling fresh cardamom immediately with the expected profit from drying and selling later. By considering the fresh-to-dried weight conversion ratio and processing costs, the system generates a practical “Hold or Sell” recommendation for farmers.

Based on the reviewed literature, most existing systems focus on individual agricultural tasks such as plant disease detection, crop grading, or price prediction. However, there is limited research on an integrated cardamom-specific platform that combines leaf disease detection, pod monitoring, pod quality grading, and market profit analysis. Existing plant disease detection systems often focus on general crops or controlled laboratory datasets, while grading systems may lack transparency and market forecasting systems may not provide farmer-specific decision support. Therefore, there is a clear need for a unified system that uses efficient deep learning and predictive modelling techniques to support Sri Lankan cardamom farmers.

The proposed Cardamo system addresses this gap by integrating EfficientNetB0-based image classification for leaf disease detection and pod monitoring, Keras-based CNN classification for pod quality grading, with Ridge Regression-based market analysis and profit

optimization. This combination provides a practical, user-friendly, and data-driven smart agriculture solution. By supporting early disease identification, pod condition monitoring, consistent quality grading, and “Hold or Sell” market recommendations, Cardamo contributes to reducing crop losses, improving export-quality production, and increasing farmer profitability in the Sri Lankan cardamom industry.

### III. METHODOLOGY

The implementation of Cardamo follows a structured methodology that combines image processing, deep learning, machine learning, and predictive analytics to support smart cardamom cultivation management. The methodology explains the steps and techniques used to develop Cardamo, an ML-driven system for cardamom leaf disease detection, pod monitoring, pod quality grading, and market-based decision support. The major stages of the methodology include data collection, data preprocessing, model development, system development, evaluation and validation, and technology stack selection.

#### A. Data Collection

TABLE I. DATA COMPOSITION

| Data Type                              | Purpose                                             |
|----------------------------------------|-----------------------------------------------------|
| Cardamom leaf images                   | Leaf disease detection                              |
| Cardamom pod images                    | Pod disease and maturity detection                  |
| Harvested cardamom pod images          | Pod quality grading                                 |
| Historical cardamom market price data  | Market price prediction                             |
| Exchange rate and processing cost data | Profit comparison and “Hold or Sell” recommendation |

Data collection was carried out according to the requirements of each component in the Cardamo system. For the leaf disease detection component, cardamom leaf images were collected to represent healthy leaves and disease-affected leaves. The disease categories include leaf blight, leaf spot. These images were used to train the deep learning model to identify disease-related visual symptoms such as discoloration, spots, texture changes, and damaged leaf regions.

For the pod monitoring component, cardamom pod images were collected to represent different pod conditions and maturity stages. These include healthy

Pods, immature pods, mature pods, capsule borer affected pods, thrips affected pods, capsule rot, pod splitting, and shrivelling. These images support the identification of maturity-related and disease-related pod conditions.

For the pod quality grading component, harvested cardamom pod images were collected to represent different quality levels. The quality levels were identified based on visible features such as pod size, color, shape, surface texture, and surface defects. These images were used to train the grading model to classify pods into suitable quality categories.

For the market analysis component, historical cardamom market price data, seasonal production information, exchange rate data, and processing cost details were collected. Processing costs include drying cost, labor cost, fuel or firewood cost, transport cost, and other post-harvest expenses. These data were used to support market price prediction and to compare the profitability of selling fresh cardamom pods immediately against drying and selling them later.

### B. Data Preprocessing

Data preprocessing was an important stage to improve the quality, consistency, and usability of the collected data. For image-based components, preprocessing was applied before model training. Leaf images, pod images, and harvested pod images were resized into fixed input dimensions according to the requirements of the selected models. Image normalization was applied to scale pixel values and improve training stability. Noise reduction and image enhancement techniques were used to reduce unwanted disturbances caused by lighting variation, camera quality, shadows, and complex plantation backgrounds.

For leaf disease detection, preprocessing improved the visibility of disease symptoms such as spots, blight regions, color changes, and damaged areas. For pod monitoring, preprocessing helped highlight pod maturity indicators and disease-related features such as color variation, surface damage, and texture differences. For pod quality grading, preprocessing supported the extraction of important visual features such as pod size, shape, color uniformity, and surface defects.

Data augmentation techniques such as rotation, flipping, zooming, brightness adjustment, and contrast variation were applied to increase dataset diversity.

This helped improve model generalization under real field conditions where images may be captured under different lighting conditions, angles, and backgrounds.

For the market analysis component, numerical data preprocessing was performed. Missing values were handled, inconsistent records were removed, and numerical features were organized into a structured format suitable for regression modelling. Historical prices, exchange rates, seasonal data, drying costs, labor costs, and other processing expenses were prepared as input features. The fresh-to-dried cardamom conversion ratio was also considered to support fair profit comparison between fresh selling and dried selling options.

The profit comparison process can be represented as follows:

$$\text{Fresh Profit} = \text{Fresh Quantity} \times \text{Fresh Market Price}$$

$$\text{Dried Quantity} = \text{Fresh Quantity} / \text{Conversion Ratio}$$

$$\text{Dried Profit} = (\text{Dried Quantity} \times \text{Predicted Dried Price}) - \text{Processing Cost}$$

### C. Model Development

TABLE II. MODELS AND APPLICATION

| Model            | Application                                     |
|------------------|-------------------------------------------------|
| EfficientNetB0   | Cardamom leaf disease detection                 |
| EfficientNetB0   | Cardamom pod disease and maturity detection     |
| Keras-based CNN  | Cardamom pod quality grading                    |
| Ridge Regression | Market price prediction and profit optimization |

The model development phase focused on developing separate models for the four major components of the Cardamo system. For the leaf disease detection component, EfficientNetB0 was used as the main deep learning model. EfficientNetB0 was selected because it provides a good balance between classification accuracy and computational efficiency. The model was trained using preprocessed cardamom leaf images to classify leaves into healthy and disease categories. The output of this component includes the predicted disease class and confidence score.

For the pod monitoring component, EfficientNetB0 was also used to classify pod images according to disease condition and maturity status. The model learns visual features such as pod color, surface texture, shape, maturity appearance, and visible damage indicators. This component supports the identification of pod conditions such as healthy, immature, mature, capsule borer affected, thrips affected, capsule rot, pod splitting, and shrivelling. The output helps farmers understand whether the pods are healthy, damaged, or suitable for harvesting.

For the pod quality grading component, a Keras-based Convolutional Neural Network model was developed to classify harvested cardamom pods into quality grades. Keras was used because it provides a flexible and efficient framework for building deep learning image classification models. The grading model analyzes visual characteristics such as pod size, color uniformity, shape, surface texture, and visible defects. This component aims to reduce subjectivity in manual grading and support farmers, traders, and exporters in making more consistent quality assessments.

For the market analysis component, Ridge Regression was used as the predictive model. Ridge Regression was selected because it is suitable for regression tasks involving multiple input variables and helps reduce overfitting through regularization. The model uses input features such as historical market prices, seasonal trends, exchange rates, drying costs, labor costs, and processing expenses to support price prediction. The prediction output is then used by the profit optimization logic to generate a “Hold or Sell” recommendation.

#### D. System Development

The Cardamo system integrates all developed models into a single smart cardamom cultivation management platform. The system is designed to support farmers through a web or mobile-based interface. Users can upload cardamom leaf images, pod images, or harvested pod images depending on the selected function. Users can also enter market-related details such as harvest quantity, current fresh price, drying cost, labor cost, and other processing expenses.

When a leaf image is uploaded, the system applies preprocessing and sends the image to the EfficientNetB0-based leaf disease detection model. The predicted disease type and confidence score are

returned to the user. When a pod image is uploaded for monitoring, the image is processed and passed to the EfficientNetB0-based pod monitoring model, which identifies disease condition or maturity status. When a harvested pod image is uploaded for grading, the Keras-based CNN model classifies the pod into the relevant quality grade.

For market analysis, the farmer enters the required market and processing details through the interface. The Ridge Regression model supports market price prediction using historical and economic input features. The profit optimization logic then compares fresh-selling profit with dried-selling profit and generates a recommendation such as “Sell Fresh” or “Hold and Dry.” Prediction results, uploaded image details, market inputs, and recommendation history are stored in a centralized database for future reference and analysis.

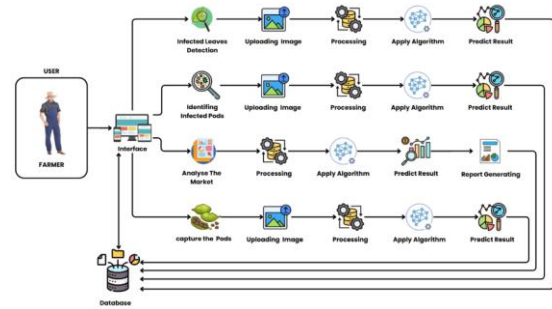


Fig. 1. System architecture diagram

#### E. Evaluation and Validation

The evaluation of Cardamo was carried out separately for each component. For the image-based components, the datasets were divided into training, validation, and testing sets. The training set was used to train the models, the validation set was used to tune performance, and the testing set was used to evaluate final model performance using unseen data.

The leaf disease detection, pod monitoring, and pod quality grading components were evaluated using standard classification metrics such as accuracy, precision, recall, and F1-score. Accuracy measures the overall correctness of predictions. Precision measures how many predicted positive results are actually correct. Recall measures the ability of the model to identify all relevant cases. F1-score provides a balanced measurement between precision and recall. Confusion matrices were also used to identify

misclassification patterns between different disease, pod condition, maturity, and quality classes.

For the market analysis component, the Ridge Regression model was evaluated using regression metrics such as Mean Absolute Error, Mean Squared Error, Root Mean Squared Error, and R-squared score. These metrics were used to measure how close the predicted values were to the actual market values. In addition, the “Hold or Sell” recommendation logic was validated by comparing generated recommendations with historical market behavior and expected profit outcomes.

Qualitative validation was also performed by reviewing selected outputs from the system. For image-based components, predicted labels were compared with visible leaf and pod conditions. For market analysis, sample farmer inputs were tested to check whether the system generated reasonable and understandable profit recommendations. This combined evaluation ensured that Cardamo was validated both technically and practically.

#### F. Technology Stack

The Cardamo system was implemented using a modern and scalable technology stack to support image processing, model training, backend development, database management, and user interaction. Python was used as the main programming language for machine learning development and backend logic. TensorFlow and Keras were used to develop the EfficientNetB0 models and the Keras-based CNN model. Scikit-learn was used to develop the Ridge Regression model for market price prediction. OpenCV was used for image preprocessing tasks such as resizing, normalization, noise reduction, and image enhancement.

FastAPI was used to build the backend API, allowing the frontend application to communicate with the trained models. PostgreSQL or Firebase was used as the database to store uploaded image details, prediction outputs, user inputs, and market analysis records. A web or mobile-based frontend was used to allow farmers to upload images, enter market details, and view results in a user-friendly format.

TABLE III. SUMMARY OF TECHNOLOGIES USED

| Technology | Purpose |
|------------|---------|
|------------|---------|

|                               |                                                               |
|-------------------------------|---------------------------------------------------------------|
| Python                        | Backend development and machine learning implementation       |
| TensorFlow/Keras              | EfficientNetB0 implementation and Keras-based CNN development |
| Scikit-learn                  | Ridge Regression model development                            |
| OpenCV                        | Image preprocessing and enhancement                           |
| FastAPI                       | Backend API development                                       |
| PostgreSQL/Firebase           | Data storage and result management                            |
| Reactnative/WebView Framework | User interface development                                    |
| NumPy/Pandas                  | Data handling, preprocessing, and analysis                    |

## IV. RESULTS AND DISCUSSION

The Cardamo system demonstrated the potential of integrating deep learning, image processing, and predictive analytics into a single smart cardamom cultivation management platform. By combining leaf disease detection, pod monitoring, pod quality grading, and market analysis, the system provides a practical decision-support solution for Sri Lankan cardamom farmers. The results and observations from the developed components indicate that the system can support early identification of crop issues, reduce manual decision-making errors, and improve cultivation and post-harvest management.

In the preprocessing stage, image resizing, normalization, noise reduction, and image enhancement were applied to improve the quality and consistency of input images. These preprocessing steps were important for handling real-world agricultural image challenges such as uneven lighting, background noise, shadows, and camera quality differences. For leaf images, preprocessing improved the visibility of disease symptoms such as leaf spots, blight regions, discoloration, and texture changes. For pod images, preprocessing supported the identification of maturity indicators, surface damage, color variation, and other visual features required for monitoring and grading.

For the leaf disease detection component, EfficientNetB0 was used to classify cardamom leaf images into healthy and disease-affected categories. The model was able to learn disease-related visual patterns from preprocessed leaf images, including symptoms of leaf blight, leaf spot, mosaic disease, chlorotic streak, and vein clearing. The use of

EfficientNetB0 provided an effective balance between accuracy and computational efficiency, making it suitable for a practical web or mobile-supported agricultural system. The output of this component helps farmers identify diseases earlier and take timely actions to reduce crop loss.

For the pod monitoring component, EfficientNetB0 was applied to classify pod conditions and maturity-related states. The model analyzed visual characteristics such as pod color, shape, surface texture, maturity appearance, and visible damage indicators. This component supports the identification of pod-related issues such as capsule borer damage, thrips damage, capsule rot, pod splitting, and shrivelling. The results showed that image-based pod monitoring can support farmers in identifying whether pods are healthy, damaged, immature, mature, or suitable for harvesting.

The pod quality grading component was developed using a Keras-based CNN model. This component focused on classifying harvested cardamom pods into quality grades by analyzing visual features such as pod size, color uniformity, shape, surface quality, and visible defects. Compared with manual grading, the automated grading approach can reduce subjectivity and improve consistency in quality assessment. This is especially useful for farmers, traders, and exporters who need reliable quality evaluation before selling or exporting cardamom products.

For the market analysis component, Ridge Regression was used to support price prediction and profit optimization. The model considered input features such as historical cardamom prices, exchange rate variations, seasonal trends, drying costs, labor costs, and other processing expenses. The system compared the expected profit from selling fresh cardamom immediately with the expected profit from drying and selling later. Based on this comparison, the system generated a “Hold or Sell” recommendation. This helps farmers make more informed post-harvest decisions instead of depending only on informal trader information or personal experience.

Overall, the results indicate that Cardamo can function as an integrated smart agriculture platform for cardamom cultivation. The image-based components support crop health monitoring and quality assessment, while the market prediction component

supports financial decision-making. Although individual models focus on different tasks, their integration into one platform provides farmers with a complete decision-support workflow from disease detection to selling strategy. Future quantitative evaluation will include performance graphs, confusion matrices, classification reports, and regression error analysis to further validate the accuracy and reliability of each component.

## V. CONCLUSION

The Cardamo system presents a meaningful step toward smart cardamom cultivation management by integrating image processing, deep learning, and predictive analytics into a single decision-support platform. The system addresses major challenges faced by Sri Lankan cardamom farmers, including leaf diseases, pod-related issues, manual quality grading limitations, and market price uncertainty.

EfficientNetB0 was used for cardamom leaf disease detection and pod monitoring, enabling the system to identify disease symptoms, pod conditions, and maturity-related visual patterns from images. A Keras-based CNN model was used for pod quality grading, supporting more consistent classification of harvested pods based on visual quality features. Ridge Regression was applied for market price prediction and profit optimization, allowing the system to provide practical “Hold or Sell” recommendations by comparing fresh and dried cardamom profitability.

Through the integration of these components, Cardamo supports early disease identification, improved pod monitoring, consistent quality grading, and data-driven market decision-making. The system can help reduce crop losses, improve export-quality production, minimize manual errors, and support farmers in making more profitable decisions. Future work will focus on expanding the dataset, improving model accuracy, adding explainable AI visualizations, validating the system with real farmers, and enhancing the platform for wider use across cardamom-growing regions in Sri Lanka

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